M4 · Phagocytosis & phagocyte defects

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

- Name the four steps of leukocyte extravasation and the adhesion molecules for each step.
- List the main chemoattractants for neutrophils and their receptors.
- Match each opsonin to its phagocyte receptor (FcγRI, CR1, CR3).
- Describe the respiratory burst from oxygen to hypochlorous acid, and the other killing mechanisms.
- Link each phagocyte defect to the step that fails, the clinical clue and the lead laboratory finding.

BCG	bacillus Calmette–Guérin
CGD	chronic granulomatous disease
CHS	Chédiak–Higashi syndrome
CR1 / CR3 / CR4	complement receptor 1 / 3 / 4
fMLP	N-formylmethionyl-leucyl-phenylalanine (bacterial formyl peptide)
FPR1	formyl peptide receptor 1
FcγR	Fc receptor for IgG
G-CSF / GM-CSF	granulocyte / granulocyte-macrophage colony-stimulating factor
G6PD	glucose-6-phosphate dehydrogenase
GPCR	G protein-coupled receptor
HLH	hemophagocytic lymphohistiocytosis
HSCT	hematopoietic stem cell transplantation
IFN-γ	interferon gamma
IL	interleukin
ICAM-1 / VCAM-1	intercellular / vascular cell adhesion molecule 1
iNOS	inducible nitric oxide synthase
LAD	leukocyte adhesion deficiency
LFA-1	leukocyte function-associated antigen 1 (CD11a/CD18)
LTB4	leukotriene B4
MAC-1	macrophage-1 antigen (CD11b/CD18) = CR3
MPO	myeloperoxidase
MSMD	Mendelian susceptibility to mycobacterial disease
NADPH	nicotinamide adenine dinucleotide phosphate (reduced form)
NET	neutrophil extracellular trap
NK cell	natural killer cell
NO	nitric oxide
PAD4	peptidylarginine deiminase 4
PRR	pattern recognition receptor
PSGL-1	P-selectin glycoprotein ligand 1
ROS	reactive oxygen species
SCN	severe congenital neutropenia

SOD	superoxide dismutase
TLR	Toll-like receptor
TNF	tumour necrosis factor
VLA-4	very late antigen 4 (CD49d/CD29)

1. Neutrophils: numbers, life span, granules

An adult makes more than 1×10^{11} neutrophils per day. Each neutrophil circulates from a few hours to up to 5 days. Abbas p.14 G-CSF and GM-CSF stimulate neutrophil production. Abbas p.14 In tissues, a neutrophil lives 1–2 days. Tissue-resident macrophages can live for years. Abbas p.15

Neutrophils have two granule types. Abbas p.14

- **Specific granules:** lysozyme, collagenase, elastase (Abbas; see the source conflict below).
- **Azurophilic granules:** myeloperoxidase (MPO), defensins, cathelicidins.

Janeway states that neutrophil α-defensins are stored in "primary" granules, which fuse with phagosomes. Janeway p.46

> **Source conflict — granules**
>
> Abbas lists elastase in the specific granules and cathelicidins in the azurophilic granules. Abbas p.14 Janeway places neutrophil elastase in the azurophilic (primary) granules. Janeway p.534 Janeway also stores cathelicidin propeptides in the secondary (specific) granules. Elastase from the primary granules activates them. Janeway p.46 For an exam item on the granule location of elastase or cathelicidins, confirm the answer with the expert.

Dead and dying neutrophils are a major part of pus. Janeway p.82

Neutrophils are usually the most numerous leukocytes in infected tissue within 24–48 hours. Monocytes then replace them. Abbas p.50 Janeway states that neutrophil influx usually peaks within the first 6 hours. Janeway p.106

> **Exam pearl**
>
> Neutrophils make little or no nitric oxide (NO). Macrophages make NO after they induce iNOS. Abbas p.15 No pus at an infected site points to a neutrophil recruitment defect (see LAD below). Abbas p.488

2. Leukocyte extravasation

2.1 Adhesion molecules

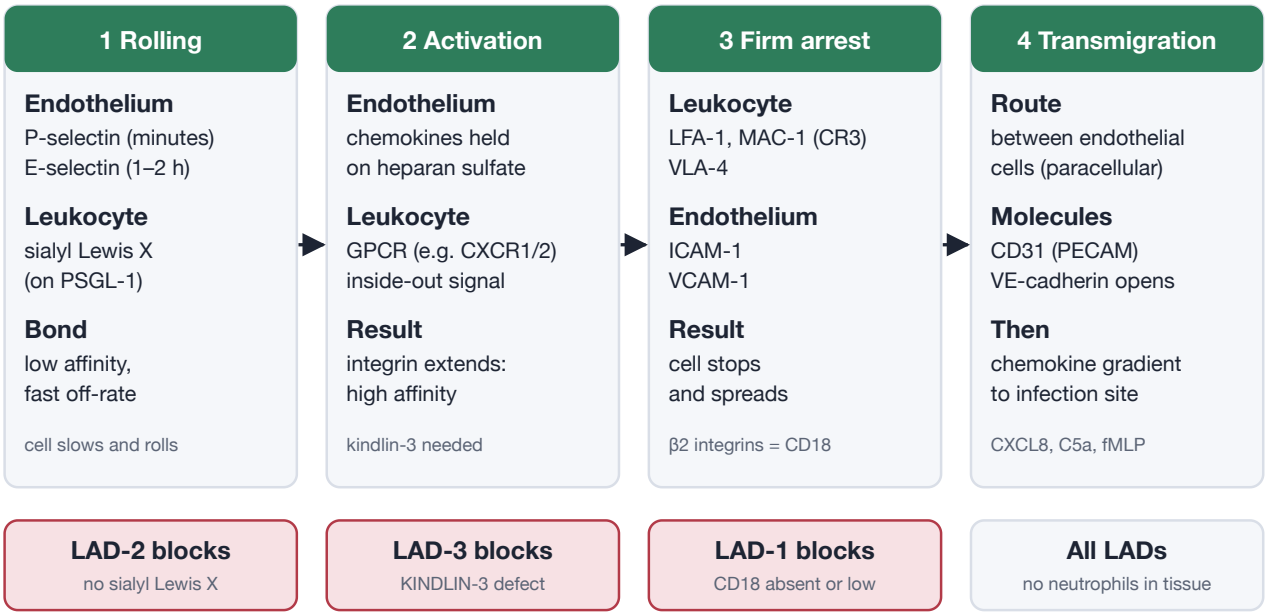
Two classes of molecules attach leukocytes to endothelium: selectins and integrins. Selectins mediate the first, low-affinity adhesion in postcapillary venules. Abbas p.44

Molecule	Where	Ligand	Source
P-selectin (CD62P)	Endothelium; stored in granules, moves rapidly to the surface after histamine or thrombin	Sialyl Lewis X on PSGL-1	Abbas p.45
E-selectin (CD62E)	Endothelium; made within 1–2 h after IL-1 or TNF	Sialyl Lewis X on glycoproteins	Abbas p.45
L-selectin (CD62L)	Leukocytes (not endothelium)	Sialomucins on endothelium	Abbas p.45
LFA-1 (CD11a/CD18)	Leukocytes	ICAM-1 (CD54)	Abbas p.45
MAC-1 = CR3 (CD11b/CD18)	Neutrophils, monocytes	ICAM-1; iC3b on opsonized particles	Abbas p.45
VLA-4 (CD49d/CD29)	Monocytes, T cells	VCAM-1 (CD106)	Abbas p.45

Integrins are heterodimers of one α chain and one β chain. Abbas p.45 LFA-1, MAC-1 and CR4 share the same β2 chain, CD18. Abbas p.309 Janeway adds that endothelial Weibel–Palade bodies hold preformed P-selectin. Janeway p.105

2.2 The four steps

1. **Rolling.** Endothelial E- and P-selectin bind sialyl Lewis X on the leukocyte. The bonds have low affinity and a fast off-rate. They form and break, so the cell rolls. Abbas p.49
2. **Integrin activation.** Endothelial heparan sulfate holds chemokines and shows them to the rolling cell. Abbas p.48 Chemokine receptor signals change the integrin shape to high affinity. This is "inside-out" signalling. Abbas p.46
3. **Firm arrest.** Cytokines increase endothelial ICAM-1 and VCAM-1. ICAM-1 binds LFA-1 and MAC-1. VCAM-1 binds VLA-4. The cell stops and spreads. Abbas p.49
4. **Transmigration.** The cell usually moves between endothelial cells (paracellular transmigration). Abbas p.49 CD31 takes part, and the VE-cadherin junction opens for a short time. Abbas p.50



D11. Leukocyte extravasation in four steps, with the LAD type that blocks each step. Abbas p.44–50 Abbas p.488

> **Exam pearl — terminology**
>
> Abbas uses "diapedesis" for the passage between endothelial cells. Abbas p.49 Janeway uses "diapedesis" for the later passage through the basement membrane. Janeway p.106 Read the question carefully. The order of events is the same in both books, but the four steps are split differently. Janeway's four steps are rolling, tight integrin binding, crossing the endothelium and basement membrane, and migration through the tissue. Janeway p.106

3. Chemotaxis

Chemotaxis is cell movement along a concentration gradient toward the source of a chemoattractant. Abbas p.46 Chemokine receptors are seven-transmembrane GPCRs. Abbas p.47 CXC chemokines with an ELR motif attract neutrophils. Abbas p.46

Chemoattractant	Receptor / target	Note	Source
CXCL8 (IL-8), CXCL1	CXCR1, CXCR2 on neutrophils	Main ELR chemokines for neutrophils	Abbas p.50
CCL2 (MCP-1)	CCR2 on classical monocytes	Monocyte recruitment	Abbas p.50
Formyl peptides (fMLP)	FPR1, a GPCR	Among the most potent leukocyte chemoattractants	Abbas p.79 Janeway p.79
C5a, C3a	GPCRs on phagocytes	Complement fragments	Abbas p.48
LTB4	GPCR on phagocytes	Arachidonic acid metabolite	Abbas p.48

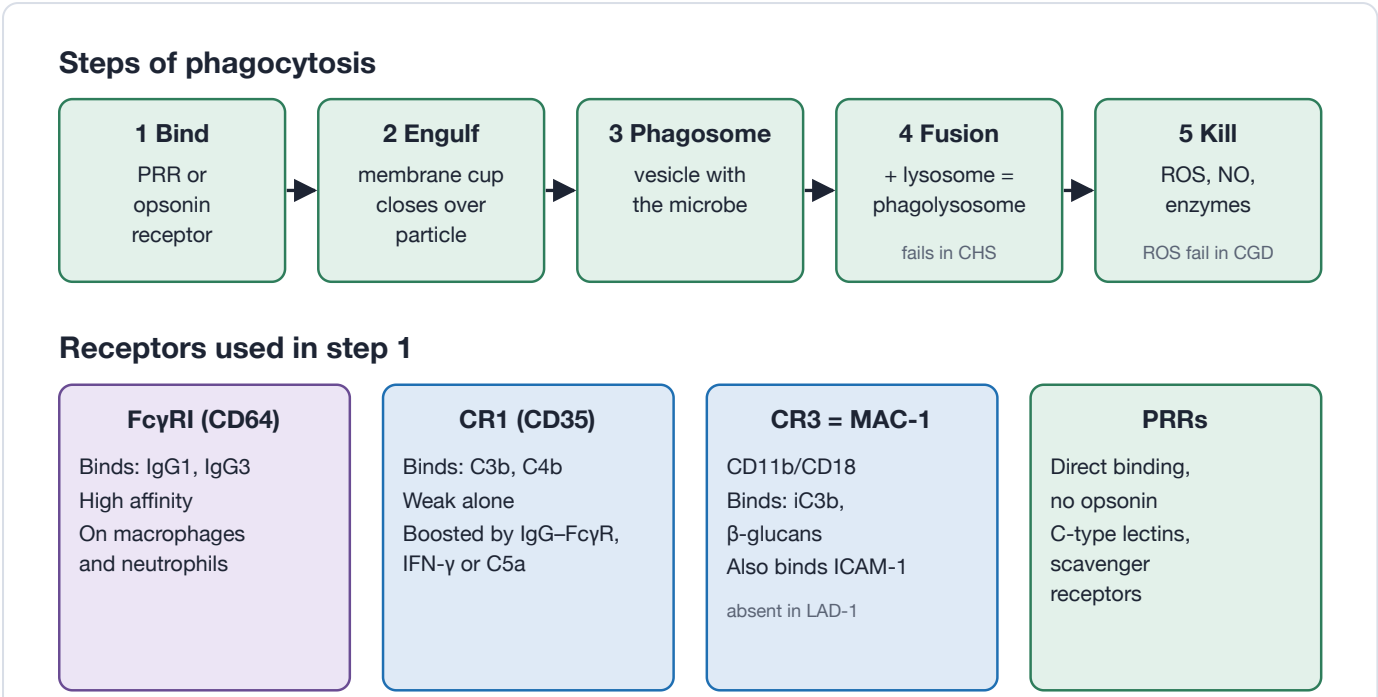
These signals also arm the cell. fMLP receptor signals induce ROS in the phagolysosome. Janeway p.79 C5a and CXCL8 activate the respiratory burst in neutrophils. Janeway p.103

4. Opsonins and phagocytosis

Opsonization is the coating of a pathogen with antibodies and/or complement proteins. Phagocytes then take it up more readily. Janeway p.49 Phagocytosis is an active, energy-dependent engulfment of particles larger than 0.5 μm. Abbas p.95

1. **Recognition.** The microbe binds a phagocyte receptor. Some receptors are PRRs, for example C-type lectins and scavenger receptors. Others bind opsonins. Abbas p.95
2. **Engulfment.** The membrane forms a cup around the particle. The cup closes and pinches off. Abbas p.95
3. **Phagosome.** The new vesicle holds the particle and fuses with a lysosome. Abbas p.95
4. **Phagolysosome.** Most microbicidal mechanisms are concentrated here. Abbas p.95
5. **Killing.** TLR, Fc receptor, C3b receptor, IFN-γ receptor and CD40 signals activate killing. Abbas p.95

Receptor	Opsonin / ligand	Key point	Source
FcγRI (CD64)	IgG1, IgG3 (high affinity)	Phagocytosis and phagocyte activation; on macrophages and neutrophils	Abbas p.297
CR1 (CD35)	C3b, C4b	Weak alone; works well with IgG–FcγR binding or IFN-γ. C5a also activates CR1 uptake.	Abbas p.307 Abbas p.313 Janeway p.63
CR3 = MAC-1 (CD11b/CD18)	iC3b; β-glucans directly	Also an adhesion integrin (binds ICAM-1); missing in LAD-1	Abbas p.309 Janeway p.78



D12. Phagocytosis in five steps (top) and the receptors that bind the microbe (bottom). Abbas p.95 Abbas p.297–313 Janeway p.63, 78 Abbas p.486; Janeway p.536

5. Killing mechanisms

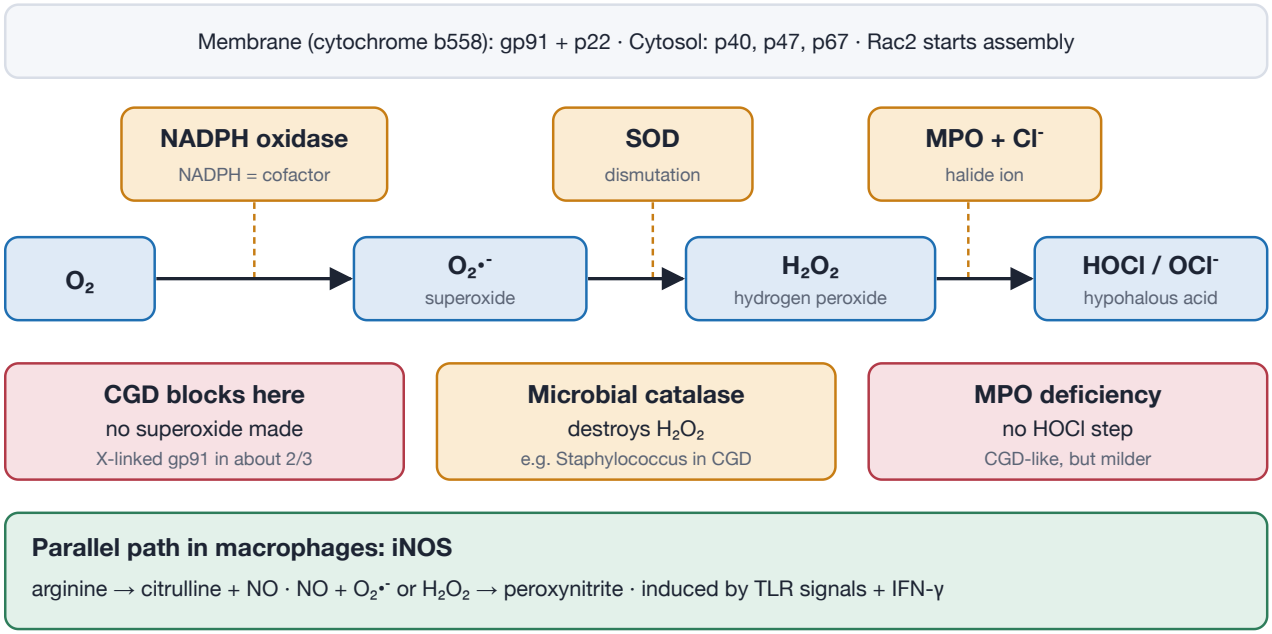
Three classes of microbicidal molecules are the most important: ROS, NO and proteolytic enzymes. Abbas p.95–96

5.1 Respiratory burst

Phagocyte oxidase (NADPH oxidase) is a multisubunit enzyme. It assembles in activated phagocytes, mainly in the phagolysosome membrane. Abbas p.96 In resting neutrophils, the cytochrome b558 subunits (gp91 and p22) sit in secondary granule membranes. The p40, p47 and p67 subunits are in the cytosol. Janeway p.81 fMLP and C5a receptor signals activate Rac2, which the assembly needs. Janeway p.80

1. NADPH oxidase reduces molecular oxygen (O_2) to superoxide ($O_2^{\bullet -}$). NADPH is the cofactor. Abbas p.96
2. Superoxide dismutase (SOD) converts superoxide to hydrogen peroxide (H_2O_2). Abbas p.96 Janeway p.81
3. MPO uses H_2O_2 to change halide ions into toxic hypohalous acids, for example hypochlorous acid (HOCl). Abbas p.96
4. Other reactions also make the hydroxyl radical. Janeway p.81

The process uses oxygen, so it is called the respiratory burst. Abbas p.96 Potassium and hydrogen ions then enter the phagolysosome. This releases and activates granule enzymes such as cathepsin G and elastase. Janeway p.81 Janeway notes that ROS alone may not kill. ROS-driven K^+ influx sets the vacuole pH for microbicidal proteins. Janeway p.536



D13. The respiratory burst chain, where CGD and MPO deficiency block it, and the iNOS path. Abbas p.96 Janeway p.80–81 Abbas p.486–487 Janeway p.536

5.2 Nitric oxide

iNOS is absent in resting macrophages. TLR signals induce it, especially together with IFN-γ. Abbas p.96 iNOS converts arginine to citrulline and releases NO. NO combines with H_2O_2 or superoxide to make peroxynitrite, which kills microbes. Abbas p.96

5.3 Enzymes and antimicrobial peptides

- **Proteases:** elastase and cathepsin G are important for bacterial killing in neutrophils. Abbas p.96
- **Defensins:** cationic peptides from epithelia and from neutrophils. They mainly disrupt microbial membranes. Abbas p.80
- **Lysozyme** digests the cell wall of some Gram-positive bacteria. **Lactoferrin** holds iron away from microbes. Janeway p.80

5.4 Neutrophil extracellular traps (NETs)

Neutrophils can extrude their DNA and granule contents. The neutrophil extracellular trap (NET) is made of DNA and histone strands with bound lysozyme, elastase and defensins. NETs trap and kill bacteria and fungi. Abbas p.96 NET formation needs PAD4 (histone citrullination), elastase, MPO and phagocyte oxidase. Abbas p.96 The neutrophil usually dies (NETosis). Excess NET formation contributes to autoimmune and inflammatory diseases. Abbas p.96

> **Clinical link**
>
> NET formation needs phagocyte oxidase and MPO. Abbas p.96 So the respiratory burst is linked to both intracellular killing and NETs.

6. Phagocyte defects

Phagocyte defects cause invasive bacterial or fungal infections of the skin, liver and respiratory tract. The fungi are mainly *Aspergillus* and *Candida*. Abbas p.486 Janeway sorts the defects into four groups: production, adhesion, activation and killing. Janeway p.533

> **Analogy**
>
> Think of a fire brigade. Neutropenia: no fire fighters. LAD: the fire truck cannot stop at the burning house. CHS: the hoses do not connect to the water tank. CGD: the water tank is empty.

6.1 Chronic granulomatous disease (CGD)

- **Cause:** mutations in components of NADPH oxidase. Abbas p.486
- **Genetics:** about two-thirds are X-linked recessive. The X-linked gene encodes gp91, the 91-kD subunit of cytochrome b558. The rest are autosomal recessive. Abbas p.486 Autosomal forms can be milder, with later onset. Janeway p.82
- **Frequency:** about 1 in 200,000 in the United States. Abbas p.486
- **Organisms:** catalase-producing bacteria such as *Staphylococcus*. Catalase destroys the H_2O_2 that host cells could make from residual superoxide. Abbas p.487 Invasive *Aspergillus* infection is the leading cause of death. Abbas p.487
- **Granulomas:** uncontrolled infection drives T cell-mediated macrophage activation. This forms granulomas and gives the disease its name. Abbas p.487
- **Complication:** secondary HLH. Abbas p.498
- **Management:** antibiotic and antifungal prophylaxis, IFN-γ, and consideration of HSCT; gene therapy trials are ongoing. Abbas p.487 After HSCT, neutrophil superoxide production of at least 15% of normal greatly improves resistance to infection. Abbas p.487

> **Check current guidance**
>
> Treatment statements come from Abbas 11th ed. Check them against current guidelines. The named diagnostic assays for CGD (dihydrorhodamine flow test, nitroblue tetrazolium test) are **not covered in provided sources — needs expert input.** The sources only state that CGD phagocytes make too little superoxide to kill ingested microbes. Abbas p.486–487

6.2 Myeloperoxidase (MPO) deficiency

MPO deficiency impairs intracellular killing. The phenotype is like CGD but less severe. Janeway p.536 Janeway lists G6PD deficiency, MPO deficiency and CGD together as defects of the respiratory burst. Janeway p.534

6.3 Leukocyte adhesion deficiencies (LAD)

All LADs are autosomal recessive. Neutrophils fail to reach the infection. Patients have severe bacterial infections from early life and cannot make pus. Abbas p.488 The infections resist antibiotic treatment, and wounds heal poorly. Janeway p.535

Type	Defect	Step blocked	Extra clue	Source
LAD-1	CD18 mutations: β2 integrins (LFA-1, MAC-1, p150,95) absent or low	Firm adhesion; also chemotaxis, phagocytosis, cytotoxicity	Leukocytosis; poor wound healing; bacterial and fungal infections	Abbas p.488
LAD-2	GDP-fucose transporter: no sialyl Lewis X	Rolling	Severe developmental abnormalities; Bombay blood group; a congenital disorder of glycosylation	Abbas p.488
LAD-3	KINDLIN-3: no inside-out integrin activation	Chemokine-induced integrin activation	Life-threatening bleeding with a normal platelet count	Abbas p.488

> **Gap**
>
> Delayed separation of the umbilical cord is a classic LAD-1 clue in teaching. It is **not covered in provided sources — needs expert input.**

6.4 Chédiak–Higashi syndrome (CHS)

- Autosomal recessive. Recurrent pyogenic infections, partial oculocutaneous albinism, giant lysosomes in neutrophils, monocytes and lymphocytes. Abbas p.488
- Gene: **LYST**, which regulates lysosome trafficking. Abbas p.488
- Phagosomes and lysosomes do not fuse properly. Phagocytes have enlarged granules and kill poorly. Janeway p.536
- NK cell function is impaired. Leukocytes also have defective chemotaxis and phagocytosis. Abbas p.488
- Metallic silver hair, poor vision, platelet dysfunction with bleeding. Janeway p.540
- Without transplantation, HLH usually develops: fever, disseminated intravascular coagulation, organomegaly, hemophagocytosis. Abbas p.488

6.5 Congenital neutropenias

- **SCN (Kostmann disease):** neutrophil count persistently below 0.5×10^9/L. Janeway p.533
- **Cyclic neutropenia:** counts swing from near normal to very low, with a cycle of about 21 days. Janeway p.533
- **Genes:** ELA2 (neutrophil elastase) is the most common cause of SCN. Some ELA2 mutations cause cyclic neutropenia. Janeway p.534 HAX1 deficiency gives an autosomal recessive SCN. Janeway p.534

- **Risk and treatment:** higher risk of myelodysplasia or myeloid leukemia; G-CSF is the mainstay of treatment. Janeway p.534

6.6 Mendelian susceptibility to mycobacterial disease (MSMD)

MSMD is severe disease from weakly virulent mycobacteria and other intracellular pathogens such as *Salmonella*. BCG vaccine can disseminate. Abbas p.498 IL-12 from dendritic cells and macrophages drives IFN-γ production by T cells and NK cells. Abbas p.498 Mutations hit IL-12p40, IL-12Rβ1, both IFN-γ receptor chains, STAT1 or NEMO (IKKγ). Abbas p.498 IL-12 and IL-23 share p40, so p40 deficiency removes both cytokines. Janeway p.538 Total absence of the IFN-γ receptor markedly reduces granuloma formation. Janeway p.538

> **Differential pearl**
>
> MyD88 or IRAK4 deficiency gives severe pyogenic infections with little inflammation ("cold" infections). Janeway p.535 This is a TLR signalling defect, not a phagocyte killing defect.

6.7 Mapping table: defect → step blocked → clinical clue → lead finding

Flow cytometry can show the absence of a cell-surface or intracellular protein. Abbas p.486 Genetic testing is often used early in the diagnosis of inborn errors of immunity. Abbas p.485

Defect (gene)	Step blocked	Clinical clue	Lead finding / test (from sources)
SCN (ELA2, HAX1); cyclic (ELA2)	Production	Pyogenic infections; cyclic: about every 21 days	Neutrophils <0.5 × 10^9/L; serial counts for cycling Janeway p.533–534
LAD-1 (CD18)	Firm adhesion	No pus, leukocytosis, poor wound healing	Flow cytometry: β2 integrins absent or low Abbas p.486–488
LAD-2 (GDP-fucose transporter)	Rolling	As LAD-1 plus developmental delay	No sialyl Lewis X; Bombay blood group Abbas p.488
LAD-3 (KINDLIN-3)	Integrin activation	As LAD-1 plus severe bleeding	Bleeding with normal platelet count Abbas p.488
CHS (LYST)	Phagosome–lysosome fusion	Partial albinism, silver hair, pyogenic infections, HLH	Giant lysosomes in leukocytes Abbas p.488; Janeway p.536, 540
CGD (gp91 X-linked; others AR)	Respiratory burst	Catalase-positive bacteria, *Aspergillus*, granulomas	Low neutrophil superoxide production; named assays: needs expert input Abbas p.486–487
MPO deficiency	H_2O_2 → HOCl	CGD-like, milder	Not covered in provided sources — needs expert input Janeway p.536
MSMD (IL-12/IFN-γ axis)	Macrophage activation by IFN-γ	BCG, environmental mycobacteria, *Salmonella*	Genetic testing of the IL-12/IFN-γ genes Abbas p.485, 498

Defect	Step blocked	Clinical clue	Lead finding
SCN / cyclic ELA2, HAX1	Production	pyogenic infections; cyclic: ~21-day cycle	neutrophils <0.5 × 10^9/L serial counts
LAD-1 CD18	Firm adhesion	no pus, leukocytosis, poor wound healing	flow cytometry: β2 integrins low
LAD-2 GDP-fucose transporter	Rolling	as LAD-1 + developmental delay	no sialyl Lewis X; Bombay blood group
LAD-3 KINDLIN-3	Integrin activation	as LAD-1 + severe bleeding	bleeding with normal platelet count
CHS LYST	Phagosome– lysosome fusion	partial albinism, silver hair, infections, HLH	giant lysosomes in leukocytes
CGD gp91 (X), others AR	Respiratory burst	catalase+ bacteria, Aspergillus, granulomas	low neutrophil superoxide output
MPO def.	H_2O_2 → HOCl	CGD-like, milder	not in sources
MSMD	IL-12/IFN-γ axis	BCG, mycobacteria, Salmonella	genetic testing

D14. Phagocyte defect map. Rows are grouped by colour: production (grey), adhesion (red), fusion (grey), killing (amber), macrophage activation (purple). Abbas p.486–488 Abbas p.485, 498 Janeway p.533–540

5 key points

1. Extravasation has four steps: selectin rolling (sialyl Lewis X), chemokine-driven integrin activation, integrin–ICAM-1 arrest, transmigration. Abbas p.49
2. CXCL8 (via CXCR1/2), C5a, LTB4 and formyl peptides attract neutrophils; CCL2 (via CCR2) attracts monocytes. Abbas p.50 Abbas p.48
3. FcγRI binds IgG; CR1 binds C3b and C4b; CR3 (MAC-1) binds iC3b and is also an adhesion integrin. Abbas p.297–309
4. NADPH oxidase → superoxide → (SOD) H_2O_2 → (MPO) hypohalous acid. CGD blocks the first step; about two-thirds are X-linked gp91. Abbas p.96 Abbas p.486
5. LAD-1 = CD18, LAD-2 = no sialyl Lewis X, LAD-3 = KINDLIN-3 with bleeding; CHS = LYST with giant lysosomes and albinism. Abbas p.488

3 exam traps

1. LAD-2 is a rolling defect (selectin ligand), not an integrin defect. LAD-1 and LAD-3 affect integrins: LAD-1 lacks the protein, LAD-3 cannot activate it. Abbas p.488
2. MPO is not the enzyme that makes superoxide or H_2O_2. NADPH oxidase makes superoxide, SOD makes H_2O_2, MPO makes hypohalous acid. Abbas p.96

3. In LAD, the blood neutrophil count is often high (leukocytosis), but tissue has no pus. Do not confuse this with neutropenia. Abbas p.488

Practice: M4 questions in the quiz (IDs M4-A1 … M4-K3).

Sources: Abbas, Cellular and Molecular Immunology, 11th ed., p.14–15, 44–50, 79–80, 95–96, 297, 307–313, 485–488, 498; Janeway's Immunobiology, 8th ed., p.46, 49, 63, 78–82, 103–106, 533–540 · Fact IDs refer to `ledger/facts.csv`.